

What's New in Pixar's Presto: ML Posing, Invertible Rigs, and Interactive Motion Blur

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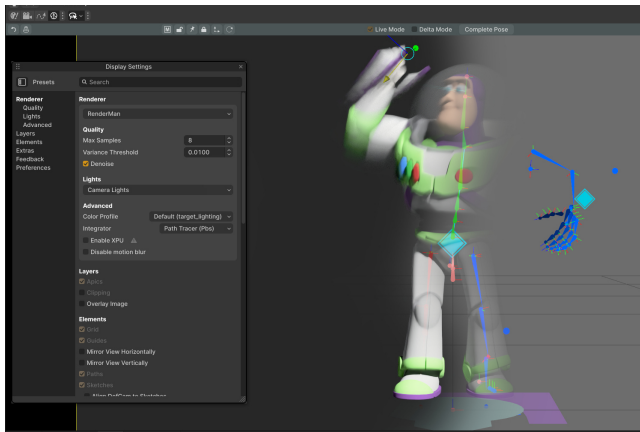


Figure 1: What's New in Pixar's Presto: ML Posing, Invertible Rigs, and Interactive Motion Blur

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1 Abstract

This presentation features three distinct technologies that together will demonstrate a unique animation experience to the SIGGRAPH Asia audience. Our ML Posing technology is the realization of the SIGGRAPH Asia 2023 paper “Pose and Skeleton-aware Neural IK for Pose and Motion Editing” running live in a production animation setting. We'll be demonstrating on Buzz Lightyear, and as we manipulate the ML Posing handles, the character does feel “alive” in a way that feels unique compared to traditional workflows. Less flashy, but actually far more involved, is the underlying “Invertible Rig” technology that ML Posing utilizes. This technology is many years in the making and represents a fundamentally new approach to rig construction: each rig component can run bi-directionally, thus making the animation controls and rig joints fully invertible at a fundamental level. Finally, interactive motion blur in Presto's viewport realizes a longtime animation request of visualizing motion blur as they animate.

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